# Tips & Tricks - Oracle Database

Databases - La Salle URL

## 1 Introduction

This documents aims to give the students some tips and tricks for the use of Oracle Autonomous Database, accessed using the SQL tool.

## 2 Understanding the Oracle ADB Architecture

Oracle Autonomous Database (ADB) uses a multitenant architecture, functioning as a multitenant container database. We are not going to get much into this concept, but this means our database is actually made up of smaller containers: a container is a collection of schemas, objects and related structures and it can include multiple pluggable-databases (PDBs). Usually, we will have multiple containers in the same database such as a container for sales, another for human resources, etc.

A PDB is a user-created set of schemas, objects, and related structures and it **appears logically as a separate DB**.

This is important because when we connect to the database we have to connect to a specific PDB (unless we want to connect to the CDB root itself).[1]

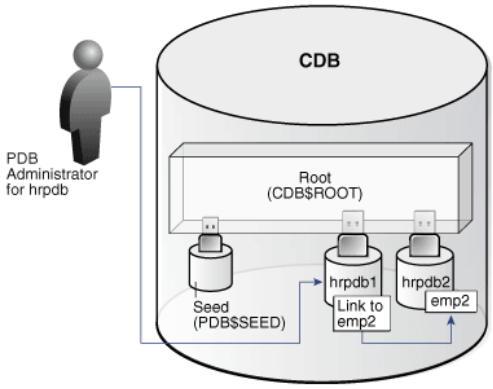
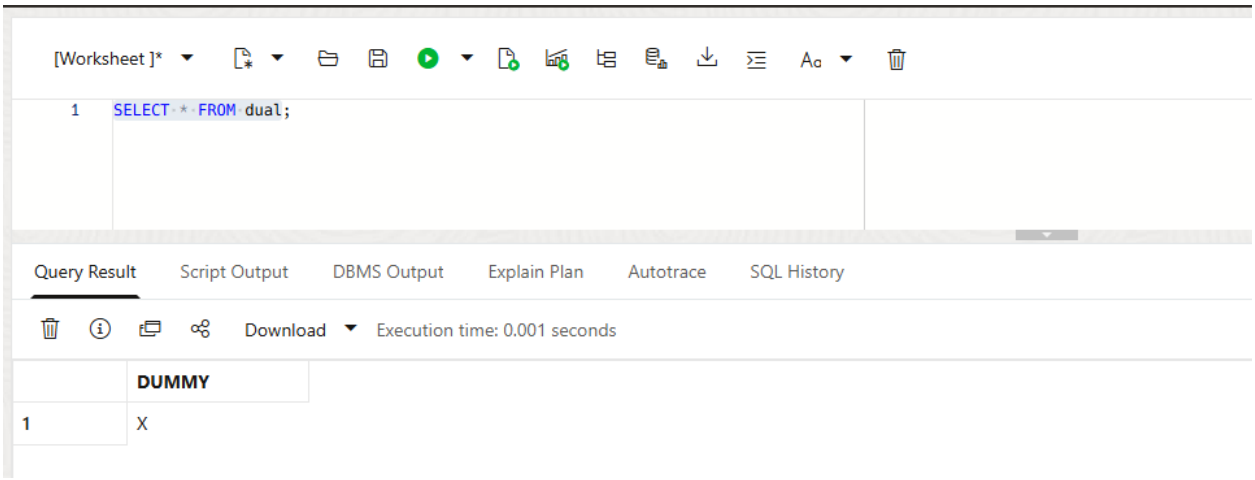



Figure 1: Multitenant Architecture - CDB[1]

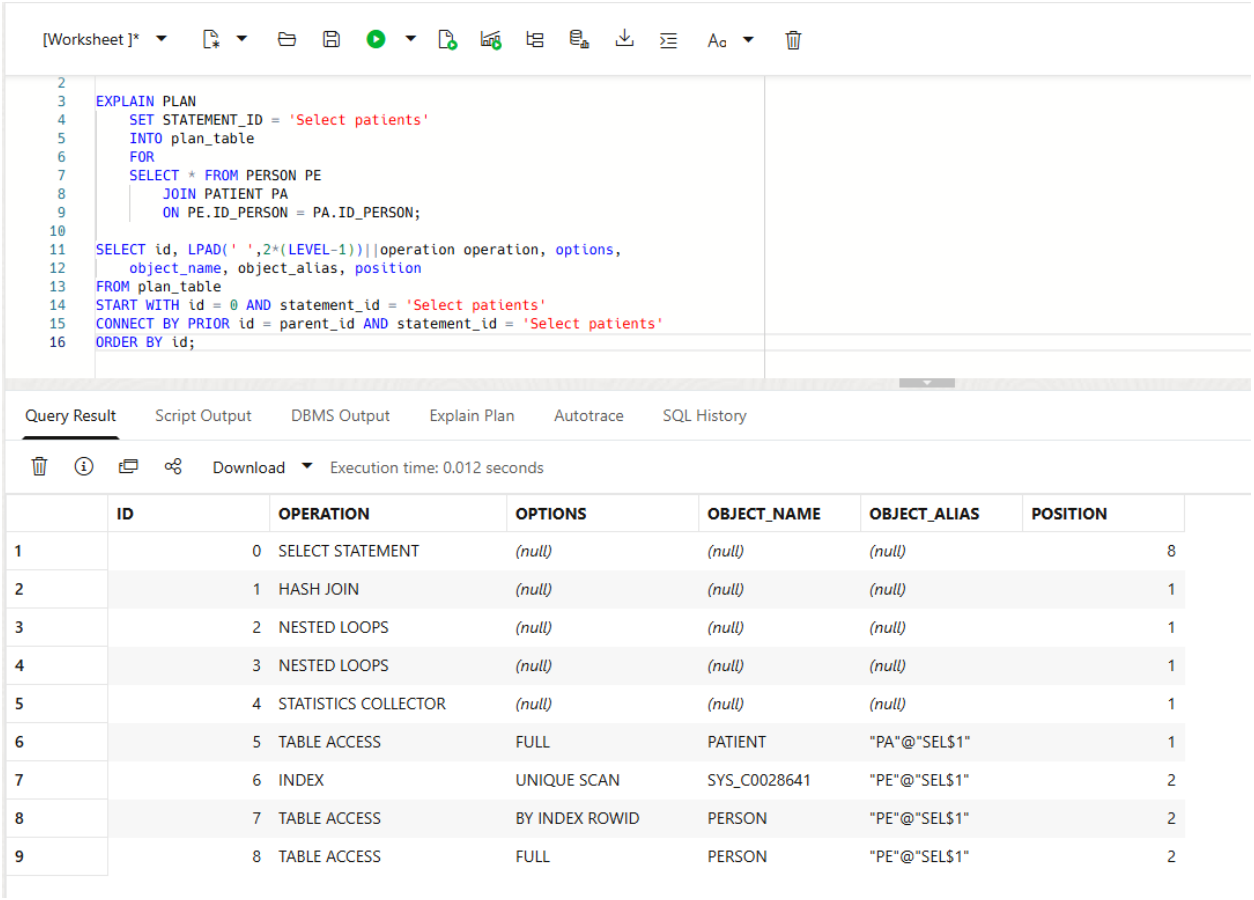
[1] Overview of Multitenant Architecture (19c)

## 3 Some tips and tricks

1. **The *dual* table:** Oracle databases include a dummy table called *dual* that can be used to test functions for example. This table has one column called *Dummy* with one single row of data (*X*):




2. **Explain plan:** To better understand the performance of a given query, the `EXPLAIN PLAN` statement aims to explain the execution plan followed by Oracle Database to execute a specified SQL statement.




| | ID | OPERATION | OPTIONS | OBJECT_NAME | OBJECT_ALIAS | POSITION |
|---|---|---|---|---|---|---|
| 1 | 0 | SELECT STATEMENT | *(null)* | *(null)* | *(null)* | 8 |
| 2 | 1 | HASH JOIN | *(null)* | *(null)* | *(null)* | 1 |
| 3 | 2 | NESTED LOOPS | *(null)* | *(null)* | *(null)* | 1 |
| 4 | 3 | NESTED LOOPS | *(null)* | *(null)* | *(null)* | 1 |
| 5 | 4 | STATISTICS COLLECTOR | *(null)* | *(null)* | *(null)* | 1 |
| 6 | 5 | TABLE ACCESS | FULL | PATIENT | "PA"@"SEL$1" | 1 |
| 7 | 6 | INDEX | UNIQUE SCAN | SYS_C0028641 | "PE"@"SEL$1" | 2 |
| 8 | 7 | TABLE ACCESS | BY INDEX ROWID | PERSON | "PE"@"SEL$1" | 2 |
| 9 | 8 | TABLE ACCESS | FULL | PERSON | "PE"@"SEL$1" | 2 |

The example in the figure is given by the following statement:

```
EXPLAIN PLAN
    SET STATEMENT_ID = 'Select patients'
    INTO plan_table
    FOR
    SELECT * FROM PERSON PE
        JOIN PATIENT PA
        ON PE.ID_PERSON = PA.ID_PERSON;

SELECT id, LPAD(' ',2*(LEVEL-1))||operation operation, options,
    object_name, object_alias, position
FROM plan_table
START WITH id = 0 AND statement_id = 'Select patients'
CONNECT BY PRIOR id = parent_id AND statement_id = 'Select patients'
ORDER BY id;
```

You can find more information about this tool here: Generating and displaying explanation plans - Oracle Database 19c

3. **Order by:** Oracle allows to order results based on the column number of the fields in the select clause:

```
SELECT
    id_person
    , firstname
    , secondname
FROM person
ORDER BY 2, 3;
```

This example will order the results by the 2nd and 3rd fields of the select clause, which are *firstname* and *secondname* respectively.

4. **Data dictionary views:** There are different metadata views in Oracle that allow us to get information of the data dictionary. It shows information for different schema objects the user has access to. The prefix of these data dictionary views can be `DBA`, `ALL` or `USER`, and depending on the prefix, we will get **all the objects in the database** (*DBA_*) [2], **the objects on which the user has some privilege** (*ALL_*), or the objects **owned by the user** (*USER_*) :

   - `<USER/ALL/DBA>_TABLES`: Table objects
   - `<USER/ALL/DBA>_TAB_COLUMNS`: Table column information
   - `<USER/ALL/DBA>_OBJECTS`: Database objects
   - `<USER/ALL/DBA>_TRIGGERS`: Trigger objects
   - ...

   You can find more information about the data dictionary views available here: Static Data Dictionary Views - Oracle 19c

5. **Dynamic performance views:**

   - The `V$NLS_PARAMETERS` table contians the value of the NLS (National Language Support) parameters, which includes the default date format, timezone format, timestamp format, etc. The value of these parameters can be set **for a specific session** using the `ALTER SESSION` statement. For example, to set the date format, you can use:

[2] We will only be able to view these views if we have the necessary privileges

```
ALTER SESSION SET NLS_DATE_FORMAT="YYYY-MM-DD HH24:MI:SS"
```

- The `V$SESSION` view holds information about the current session and can be interesting to check whether there are locks caused by active sessions or to check for performance problems.

You can find a list of all the dynamic performance views available here: Dynamic Performance Views - Oracle 19c

6. **Truncate table:** Instead of deleting all the data in a table, you can use the `TRUNCATE` clause. The truncate clause is considered a DDL clause in Oracle and therefore, it cannot be rollbacked. It allows us to avoid triggers, and removes data all at once instead of using a cursor:

```
TRUNCATE TABLE person;
```

7. **Describe a table:** You can use the `desc` or `describe` clause to get a table description:

